

ORIGINAL RESEARCH

A lightweight image splicing tampering localization method based on MobileNetV2 and SRM

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Abstract

The architectures of many state-of-the-art local tampering detection models are complexity, and the training process of those models is also time-consuming. Therefore, this paper constructs a lightweight local tampering detection method based on the convolutional network MobileNetV2 and a dual-stream network. Specifically, the algorithm first improves the MobileNetv2, which not only reduces the multiple of its downsampling operator to retain richer traces of image tampering, but also introduces the dilated convolution in it to expand the receptive field of feature maps. The dual-stream network uses RGB stream to extract image tampering features such as strong contrast difference and unnatural tampered boundaries, and implements spatial rich model (SRM) stream to extract image tampered area and noise features of real area. Finally, the features extracted from two streams are fused through an improved attention mechanism called parallel convolutional block attention module (CBAM), which can improve the sensitivity of the model to important features in RGB and SRM. The experimental results show that the proposed algorithm still has higher positioning accuracy than some existing algorithms, while achieving lightweight.

1 | INTRODUCTION

With the rapid development of information technology, mobile phones, tablets, digital cameras and other electronic devices are rapidly popularized, and the acquisition of digital images has become increasingly convenient and has become the main way and important carrier for us to acquire and transmit information. Compared with text, images can effectively and accurately convey richer information, and their intuitive and vivid characteristics are easier to attract people's attention, enrich and beautify people's visual experience. Therefore, images have been widely used in real life. However, while digital images bring convenience to human's lives, they also bring some security risks to the society.

In the early days, due to the limitations of image acquisition methods and processing methods, it's hard for ordinary people to modify the image content and cover up the tampering traces. Nowadays, digital image acquisition equipment is more and more advanced, especially with the development and popularization of image editing software such as Photoshop, ACD-See etc., which greatly reduces the cost of image tampering and forgery. Professional users can use these image

editing tools to modify the image content without leaving obvious visual traces. Although most people only use the modified pictures for their own purposes, there are also some lawbreakers who have illegally edited the content of the images and used the tampered images to achieve their own commercial, political and military purposes [1], which has caused great harm to the credibility of public utilities. Under the background of increasingly serious image tampering problems, the society urgently needs an effective and reliable technology to authenticate the authenticity of digital images. Therefore, the content protection and integrity authentication of digital images have attracted more and more researchers' attention, and researching on digital image forensics has also become one of the hotspots in the field of information security.

There are various ways of image tampering, but different tampering methods leave special traces in the image [2]. Digital image forensics technologies determine whether the image has been tampered with by analysing the noise characteristics and statistical characteristics inside the image [3]. According to whether image preprocessing is required, digital image forensics technology can be divided into two categories [2]: active forensics and passive forensics. Active forensics technologies need

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to embed key information in the image in advance, while passive forensics technologies need not. In addition, the passive forensics technologies use the content characteristics or statistical characteristics of the image itself to identify the authenticity of the image, so that it is more suitable in practice. Passive forensics techniques can be roughly divided into the following three categories [4]: source image forensics, fuzzy image forensics, and image content forensics. Source image forensics [5] judges the authenticity of the image by analysing the statistical characteristics of the image to detect whether the image is taken by a specific camera source. The fuzzy image forensics [6] is to verify the reliability of the image content by restoring the blurred image. The last category, namely the content forensics [7], is to determine whether the image has undergone image processing operations such as median filtering, resampling, and JPEG re-compression, and whether the image content has been tampered with.

Compared with the other two forensic methods, digital image content forensics is more widely used. There are roughly three tampering ways of image content [2]: image copy-paste, image splicing, and image erasure. Among them, image splicing and copy-paste are the two most common tampering scenes. The difference between them is that the source of the tampered area is different. The tampered area of image copy-paste comes from the original image itself, while the tampered area of image splicing comes from another image, and image erasing is to remove part of the original image and fill it. After the content of digital images is tampered with in the above three ways, statistical features such as colour features and pixel continuity within the image will change. Researchers can judge the authenticity of the image content according to the corresponding tampering features. After the criminals complete the tampering of the image content, they will also add additional image processing operations to the tampered image to cover up the traces of tampering. Commonly used image processing operations include median filtering [8], Gaussian smoothing [9], JPEG recompression [2] etc. For these types of image processing operations, researchers have also proposed corresponding forensic algorithms.

For image copy-paste tampering, Ardizzone et al. [10] proposed an improved block-based matching algorithm, which can accurately locate the copy-paste tampering area of an image and effectively resist post-processing operations such as geometric distortion. Yang et al. [11] proposed an image copy-paste forensics algorithm based on mixed features where an interest point detector KAZE was introduced, and the SIFT features were extracted to extract more feature points. Experimental results proved that the algorithm not only can correctly detect image tampering region, but also successfully resist image post-processing operations. Hashmi et al. [12] used robust features called SURF for feature transformation and wavelet transform to localize tampered regions of the image.

For image splicing tampering localization, noise features, CFA interpolation, JPEG compression features and other features can be used to locate the tampered area of the image in the general way. Different cameras have different CFA interpolation algorithms and unique pattern noise, which means that the different characteristics of the tampered area and the source

of the real area can be used to identify the tampered area. Zeng et al. [13] proposed a tampering region localization algorithm based on noise features. First, the image is divided into several blocks, and then the noise of each image block is graded by the principal component analysis algorithm, and finally those blocks are divided into real image patches and tampered image patches by the *K*-means algorithm. Ferrara et al. [14] proposed a tampered region detection algorithm based on CFA interpolation, which can obtain the tampering probability of each 2×2 image block without prior tampering region location. Liu et al. [15] extracted shadow boundaries and penumbra regions in images, and detected tampered regions through regular properties of light sources and inconsistencies in shadow patterns. The detection algorithm based on JPEG recompression assumes that the tampered area has undergone multiple JPEG compressions, and related algorithms [16, 17] use the traces left by multiple compressions to detect the spliced area.

In addition to the above traditional methods, in recent years, researchers have also proposed some image tampering detection methods based on deep learning networks. Zhang et al. [18] made a preliminary attempt to locate the tampered area, and constructed a multi-layer perceptron model to divide the image into 32×32 small blocks for tampering classification detection, but only some rough square areas were obtained. Bappy et al. [19] proposed a hybrid model of CNN-LSTM to predict tampered regions of an image, and used (Long Short-Term Memory) LSTM to capture the recompressed features of image patches, but the method could not identify image patches completely located within the tampered regions. Salloom et al. [20] proposed the deep neural network model called MFCN, which uses VGG-16 as the backbone network [21] to predict the tampered area and tampered boundary of the image, and then use the prediction result of the tampered boundary to correct the tampered area, so that the segmentation accuracy of the tampered area is greatly improved. Cun et al. [22] improved on the CNN-LSTM model, adding a classification network that predicts the likelihood of image patch tampering, and used a conditional random field (CRF) to improve the prediction results of tampered regions, but its positioning accuracy of the tampered area is still not good. Zhou et al. [23] proposed a dual-stream network based on Faster-RCNN [24], which first extracted tampered features such as strong contrast differences, unnatural tampered boundaries in the image through the RGB stream, and also used the SRM [25] filter. The noise difference features between real and tampered regions are extracted. In addition, there are some tampering detection methods based on image similarity [26] and semantic segmentation [27]. Similar to semantic segmentation, the difficulty of image tampering area segmentation is also to find area boundaries accurately.

When extracting image features using the above methods, the semantic information and tampering traces of the image are destroyed in the process of conducting multiple downsampling operators, resulting in the final detection performance is poor. Meanwhile, with the increasing depth and size of convolutional neural networks, the high consumption of storage space and computing resources makes it difficult to effectively apply on various hardware platforms. Overall, the compression

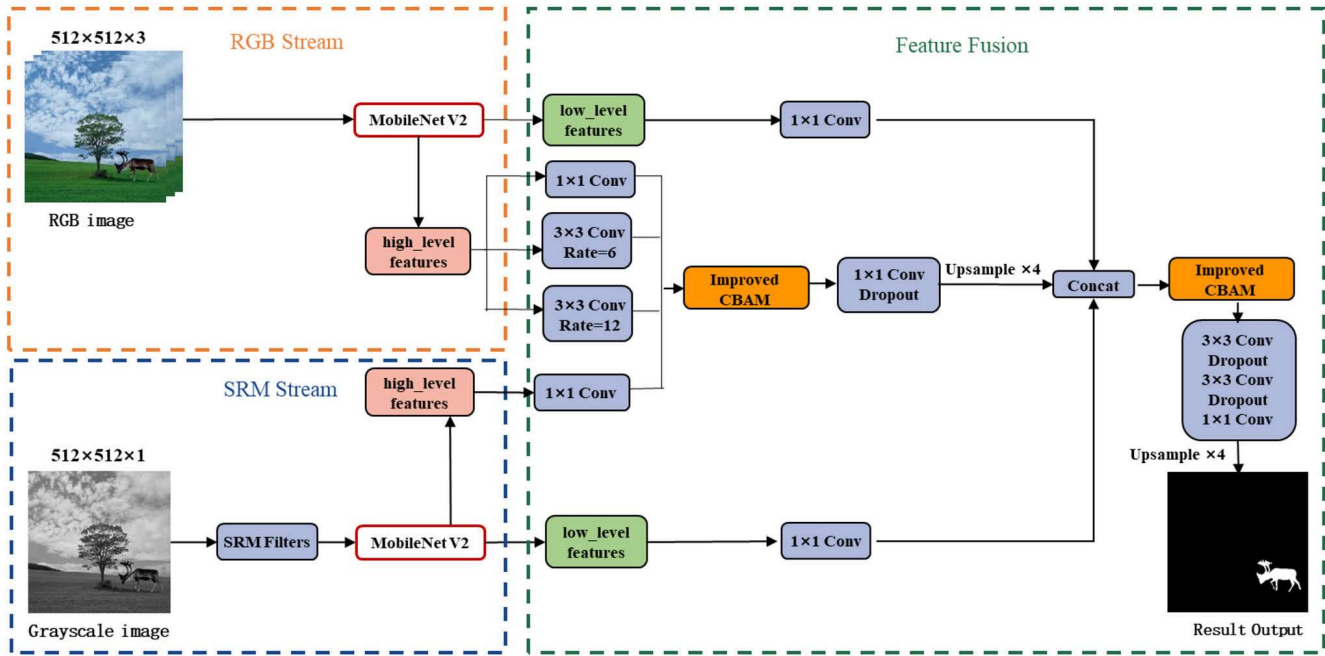


FIGURE 1 The network architecture of the proposed image splicing tampering localization method based on MobileNetV2 and SRM

and acceleration of deep learning models become necessary and urgent.

Therefore, in this paper, a lightweight image splicing and tampering localization algorithm is proposed, which is composed of a dual-stream feature extraction module and a feature fusion module. In detail, the RGB stream is used to extract the edge texture features of the image tampered area, and the SRM stream is used to extract the noise features of the tampered area and the real area, and both two streams use MobileNetV2 as backbone network to extract image features. The extracted features were then fused by an improved attention mechanism called improved CBAM, in which the stream attention mechanism and the spatial attention mechanism are parallel processing the input features rather than serial processing of those features, so that two attention mechanism do not interfere with each other that can select more discriminating features.

The rest of paper is organized as follows. Section 2 describes the proposed tampering localization algorithm in detail including the dual-stream network and the improved CBAM attention mechanism. Section 3 presents all the experimental results and also discusses the performance of the proposed algorithms. Section 4 concludes this paper.

2 | THE PROPOSED NETWORK ARCHITECTURE

The network architecture of the proposed image splicing and tampering localization method is shown in Figure 1. The model consists of RGB stream module, SRM stream module and feature fusion module. The inputs of the RGB stream and SRM stream are RGB images and grayscale images, respectively, and

both streams use MobileNetV2 as the backbone network to extract image features. We will describe these four parts, namely the backbone network MobileNetV2, the feature extraction procedures of RGB Stream and SRM Stream, and feature fusion module in detail.

2.1 | MobileNetV2

At present, convolutional neural networks have been widely used in the field of computer vision. In order to make the model achieve higher accuracy in tasks, the depth of the model continues to increase. The depth of ResNet [9] has reached 152 layers, which leads to the increasing need of storage space and computing resources for training. In this regard, researches proposed a lightweight deep neural network MobileNetV2, which aims to reduce the size of the model to improve its learning speed while maintaining the performance of the model. The comparison of MobileNetV1 and MobileNetV2 is shown in Figure 2.

2.1.1 | Depthwise separable convolution

The main idea of MobileNetV1 [28] is to use the depthwise separable convolution to replace the standard convolution, that is, to decompose the standard convolution into the depthwise convolution and the pointwise convolution, which play the role of filtering and linear combination, respectively. As shown in Figure 3, it is assumed that the size of the input feature map is $M \cdot D_F \cdot D_F$, where D_F is the length and width of the feature map, and the size of the output feature map is $D_F \times D_F \times N$. For the standard convolution $D_K \cdot D_K$, the cor-

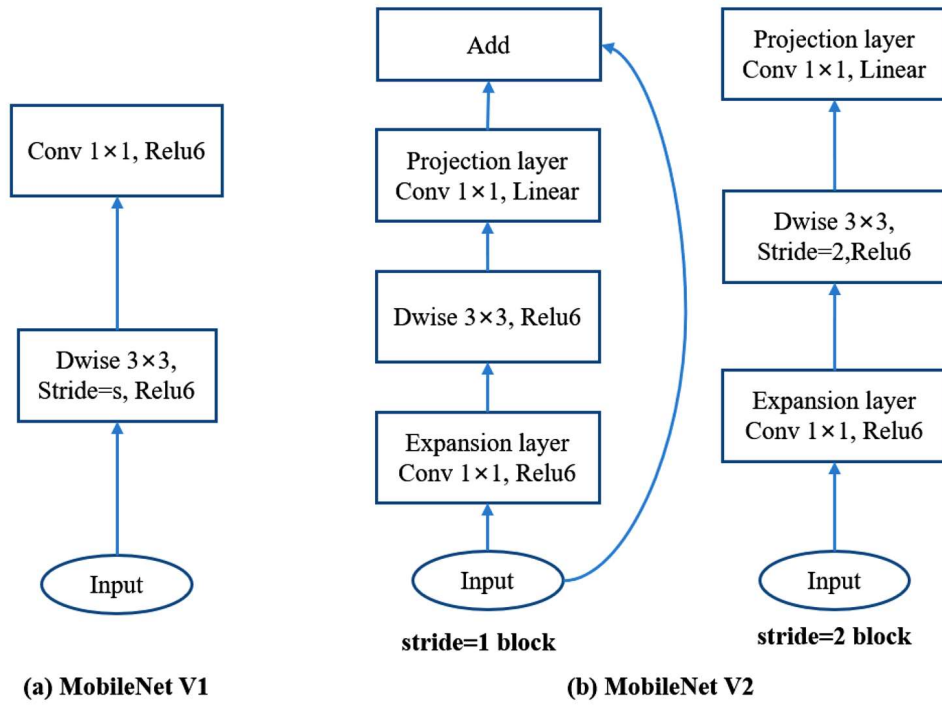


FIGURE 2 The architectures of MobileNetV1 and MobileNetV2

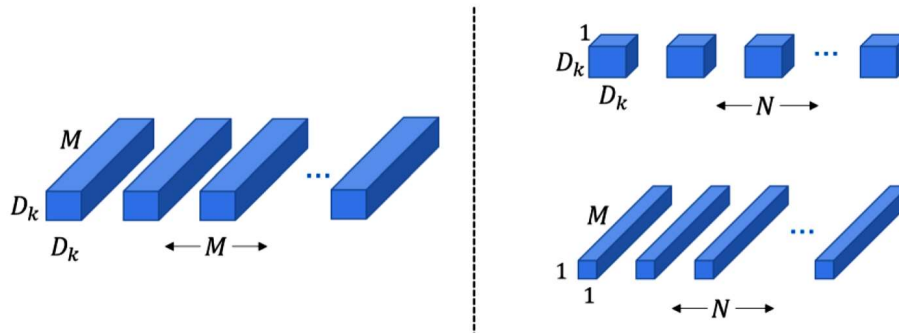


FIGURE 3 The left side of the figure is the standard convolution, and the right side of the figure is the depthwise separable convolution

responding calculation amount is $D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F$. For the depthwise separable convolution, the computation of the depthwise convolution is $D_K \cdot D_K \cdot M \cdot D_F \cdot D_F$, and the computation of the pointwise convolution is $M \cdot N \cdot D_F \cdot D_F$, so the total calculation amount of the the depthwise separable convolution is $D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F$, and its ratio to the standard convolution calculation amount is shown in Equation (1):

$$\frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_K^2}. \quad (1)$$

In general, the value of N is relatively large. When $D_K = 3$, the calculation amount of the depthwise separable convolution is about $1/9$ of that of the standard convolution.

2.1.2 | Inverted residuals

Compared with the stacked structure similar to VGG in MobileNetV1, inverted residual block [29] is used in MobileNetV2, in which shortcut connection in ResNet is implemented into the model. Figure 2a represents the convolution module used in MobileNetV1, while Figure 2b is the convolution module in MobileNetV2 where there are two cases of stride = 1 and stride = 2. The structure of the convolution module in MobileNetV2 is narrow on both sides and wide in the middle width is just the opposite of the residual block in ResNet, so it is called inverted residual block. In the inverted residual block, in addition to continuing to use the depthwise separable convolution, the expansion layer and projection layer structures are also used, both of which use a 1×1 convolution kernel. In detail, the expansion layer is responsible for mapping low-dimensional space to high-dimensional space and its default

TABLE 1 Parameters of the improved MobileNetV2 network structure

Input	Operator	T	C	N	S
512×512	Conv2d	—	32	1	2
256×256	Bottleneck	1	16	1	1
256×256	Bottleneck	6	24	2	2
128×128	Bottleneck	6	32	3	2
64×64	Bottleneck	6	64	4	2
32×32	Bottleneck	6	96	3	1
32×32	Bottleneck	6	160	3	1
32×32	Bottleneck	6	320	1	1

expansion factor is 6, while the projection layer is responsible for mapping high-dimensional space to low-dimensional space. The ReLU activation function of the last layer is replaced by a linear activation function, since using the ReLU activation function may cause information loss or destruction when converting features from high-dimensional to low-dimensional.

2.2 | RGB stream module

The RGB stream is one of the main streams in the model, which can effectively extract the edge texture features of the tampered area. The input image size is a 512×512 RGB image, and the number of streams is 3. As shown in Figure 1, the model uses the improved MobileNetV2 as the backbone network to extract image features, and its detailed structure is shown in Table 1, where 'Input' represents the input feature map size, 'Operator' is the operation module, 'Conv2d' means the standard convolution, bottleneck denotes the inverted residual block mentioned above, 'T' is the expansion multiple, 'C' is the number of output streams, 'N' is the number of modules, and 'S' is the convolution step size.

As can be seen from Table 1, this paper improves the MobileNetV2 network to maintain more image details. The original MobileNetV2 network accepts an image with a size of 224 × 224, and the downsampling multiple is 32 times. The size of the final output feature map is 7 × 7, and then passed to a global average pooling layer to get a one-dimensional feature vector. In this paper, the size of the input image is changed from 224×224 to 512×512, which further amplifies the tampered area features of the image. In addition, the downsampling factor is changed from 32 to 16, thereby preserving more image details. The extracted shallow features are the features of the fourth layer in Table 1, the size of the feature map is 128 × 128 × 24, the deep features are the features of the last layer, and the size of the feature map is 32 × 32 × 320.

2.3 | SRM stream module

The SRM stream plays an auxiliary role in the segmentation of the image tampering area. The input image is a 512×512 grayscale image, and the number of streams is 1. The module

uses 3 SRM filters, with their size are 5 × 5, 3 × 3 and 1 × 3, respectively, to extract the noise features of grayscale images. However, the later other two SRM filters also have good effect, but their size is less than 5 × 5. Therefore, we performed the zero-filling processing for other areas. The weights of these filters are defined as follows:

$$\begin{aligned}
 K_{5 \times 5} &= \begin{bmatrix} -1 & 2 & -2 & 2 & -1 \\ 2 & -6 & 8 & -6 & 2 \\ -2 & 8 & -12 & 8 & -2 \\ 2 & -6 & 8 & -6 & 2 \\ -1 & 2 & -2 & 2 & -1 \end{bmatrix}, \\
 K_{3 \times 3} &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 2 & -4 & 2 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \\
 K_{1 \times 3} &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (2)
 \end{aligned}$$

During the model training process, the weights are kept fixed that are the same as those used in identifying image processing operations. The extracted noise features are then passed through the improved MobileNetV2 network to extract shallow features and deep features, and finally fused with the features extracted from the RGB stream.

2.4 | Feature integrate module

After extracting the shallow features and deep features of the image, respectively, by the RGB stream and SRM stream, the features are then fused in the feature integrate module. Since the deep feature information obtained by the RGB stream is relatively rich, the model uses a 1×1 convolution and two dilated convolutions to further process the features. After that, the obtained features are concatenated with the deep features of the SRM stream, and then the improved convolutional block attention module (CBAM) attention mechanism is used to select more discriminative features from the concatenated features, the structure of which is shown in Figure 4.

Specifically, the input image feature F first passes through the channel attention module and the spatial attention module to obtain the attention-enhanced feature maps $M_C(F)$ and $M_S(F)$, respectively, which are added to output the final feature map $M_T(F)$. Channel attention module mainly processes the feature maps of different channels and tells the model to pay more attention to those feature maps. The spatial attention module mainly processes the feature regions of the feature map and tells the model which feature regions of the feature

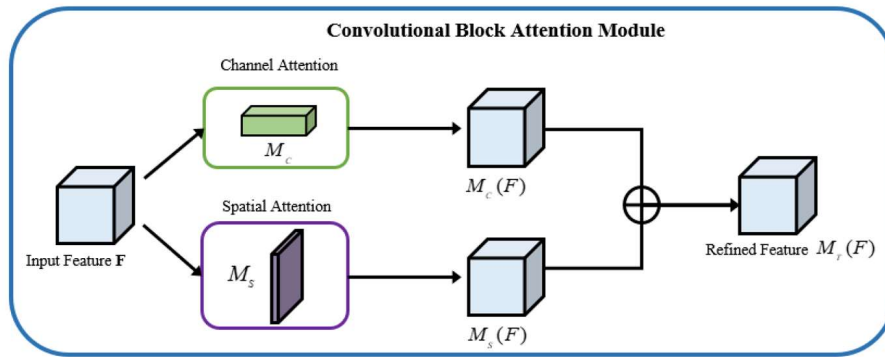


FIGURE 4 Improved CBAM attention mechanism

map should be paid more attention to [30]. The computation of $M_C(F)$ and $M_S(F)$ are shown in the Equations (3) and (4), the stream attention module and the spatial attention module are changed from the general serial to the parallel of the stream and spatial attention mechanism, and this decouple operation makes the two attention mechanisms do not interfere with each other.

$$M_C(F) = F \cdot \sigma(MLP(MaxPool(F)) + MLP(AvgPool(F))), \quad (3)$$

$$M_S(F) = M_C(F) \times \sigma(f^{7 \times 7}([MaxPool(M_C(F)); AvgPool(M_C(F))])). \quad (4)$$

After the attention mechanism, the output feature map is processed by 4 times upsampling. In order to reduce the computational cost of the model, the upsampling operation adopts the linear interpolation method, in which no additional model parameters need to be introduced. The feature map after upsampling is concatenated with the shallow features obtained by the 1×1 convolutional layer, and then the improved attention mechanism is used to integrate the multi-scale features of the RGB stream and the SRM stream. The output feature map is sub-sampled after multiple convolution operations and dropout operations, and the final output is the prediction result of the model.

3 | EXPERIMENTS

This section is including several parts: dataset, experimental environment and training process, comparative experimental results, and ablation experimental results for detailed introduction. The trained model was also tested on two datasets to verify its effectiveness.

3.1 | Datasets

The proposed lightweight model in this paper uses CASIA v2.0 as the training set, and the performance of the trained model

is evaluated on CASIA v1.0 and Columbia. CASIA v2.0 contains 7491 original images and 5123 tampered images. After data cleaning, a total of 4780 images are used as the training set. The test set is CASIA v1.0 and Columbia. There are 921 tampered images in the CASIA v1 data set, including 461 stitched tampered images and 460 copy-paste tampered images. There are 180 spliced and tampered images in the Columbia dataset, and the image format is TIF.

3.2 | Implementation details

The experimental environment configurations are: the CPU model is I7-5930k, the memory is 32 GB, and the GPU model is GeForce 1080Ti with 11GB video memory. The model is built based on the Pytorch deep learning framework, and the optimization method in the training process is stochastic gradient descent. The initial learning rate is set to 0.003, the momentum is set to 0.9, the weight decay is set to 0.0005, and a total of 80 epochs of training are performed. The loss function during training is cross entropy, which can be calculated as:

$$L_{Mask} = -\frac{1}{C} \sum_{i=0}^{C-1} (\alpha \cdot y_i \ln(y'_i) + \beta \cdot (1 - y_i) \ln(1 - y'_i)). \quad (5)$$

Among them, y_i and y'_i are the real value of each target point in the image area and the predicted value of the model, C is the total number of pixels in the received image each time, and the values of α and β are 10.0 and 1.0, respectively.

3.3 | Evaluation metrics

This paper uses F1-Score and MCC-Score as the evaluation metrics of the model, both of which can be used to evaluate the detection accuracy of tampered regions. F1-Score is the weighted harmonic average of recall and precision, and MCC-Score is a relatively balanced indicator, which can better evaluate

the model when the sample content of the two categories is very different, which is more applicable in the field of tamper detection.

F1-Score is calculated as:

$$F_1(M_{Out}, M_{GT}) = \frac{2TP}{2TP + FN + FP}. \quad (6)$$

The MCC-Score is denoted as:

$$MCC(M_{Out}, M_{GT}) = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \quad (7)$$

where M_{Out} and M_{GT} represent the prediction result of the model and the tampered area Mask of the image, respectively, TP represents the number of tampered pixels that are correctly classified, TN represents the number of real pixels that are correctly classified, FP represents the number of tampered pixels that were misclassified, and FN represents the number of tampered pixels that were incorrectly classified.

3.4 | Comparative experiment

The image splicing tampering localization algorithm proposed in this paper is compared with other benchmark algorithms, namely the algorithm proposed by Zhou et al. [23], MFCN [20], DCT [31], CFA2 [32], EXIF-SC [33], SpliceRada [33] and MSGL [34]. Here, Zhou proposed a dual stream Faster R-CNN network and trained it end-to-end to detect the tampered area of a given image. MFCN is a multi-task fully convolutional network, which uses two output branches for multi-task learning. DCT is a passive method to detect digital image forgery by measuring its quality inconsistency of blocking artefact. CFA2 has introduced tamper detection techniques based on artefacts created by colour filter array (CFA) processing in most digital cameras. EXIF-SC and SpliceRadar are two non-end-to-end deep learning-based splice localization tools, which were proposed recently and have shown good performance in image splice detection. MSGL is a deep learning model for image splicing localization that not only considers local image features, but also extracts global information of images by using a multi-scale guided learning strategy. In the experiment, the performance of all the tested algorithms are evaluated by the average F1-Score and MCC-Score scores, and the results are shown in Table 2.

As can be seen from the table, compared with all the other deep learning tampering detection algorithms, the lightweight model proposed in this paper achieves the highest detection accuracy on CASIA v1.0. It should be noted that the model can achieve an inference speed of 32 FPS for an image with a resolution of 256×384 , namely it cost 0.75 s to give an interface result of an input image. In addition, if only the RGB stream is used for tampering detection, the inference speed reaches at

TABLE 2 Experimental results of the algorithm

Methods	CASIA v1.0		Columbia	
	F1-score	MCC-score	F1-score	MCC-score
Our approach	0.7508	0.7294	0.5873	0.4316
Zhou	0.6849	NA	0.5370	NA
MFCN	0.5410	0.5201	0.6040	0.4645
DCT	0.3005	0.2516	0.5199	0.3256
CFA2	0.2125	0.1615	0.1785	0.1542
EXIF-SC	0.6195	0.5817	0.5181	0.4512
SpliceRadar	0.5946	0.5397	0.4721	0.4199
MSGL	0.6457	0.5941	0.5386	0.4278

48 FPS, namely it cost only 0.5 s to give an interface result of an input image. The F1-Score results of our methods achieved second-best among all the algorithms, and MCC-Score of the method is comparable with the other tested methods. The reason is, images in Columbia dataset are uncompressed, and the splicing traces can be easily discerned by the human eye. Models that emphasize local features are more suitable for this dataset. The detection results of the model are shown in Figure 5. The first row is the original image, the second row is the tampered image, and the third row is the detection result of the model. It can be seen that the proposed model still has a high degree of segmentation accuracy for image tampering area.

3.5 | Ablation experiment

This paper also conducts ablation experiments on the two proposed improvement points, and the detection accuracy in the CASIA v1.0 dataset using different model structures is shown in Table 3. Among them, Base model represents using only RGB streams to predict the tampered area of the image, SRM Stream represents using two streams for prediction, self-attention represents using the improved CBAM attention mechanism module.

From Table 3, it can be found that compared with the Base model, the integration of SRM Stream can effectively improve the detection accuracy of the model (i.e. F1-Score is increased by about 1.5 percentage points). This may due to the fact that the model integrates the original tampered area texture features and noise features, so that the model can accurately detect the tampered area. After the attention mechanism is implemented, the detection accuracy of the model is also greatly improved, for the F1-Score is increased by about 2.5 percentage points. Obviously, the addition of the attention mechanism can effectively enhance the sensitivity of the model to important features and reduce the interference of irrelevant features. The final model has a segmentation accuracy of 0.7653 (F1-Score) on the CASIA v1.0 dataset. Figure 6 has shown the detection results of ablation experiments, which verify the effectiveness of the improved CBAM attention mechanism and the two-stream feature extraction module.

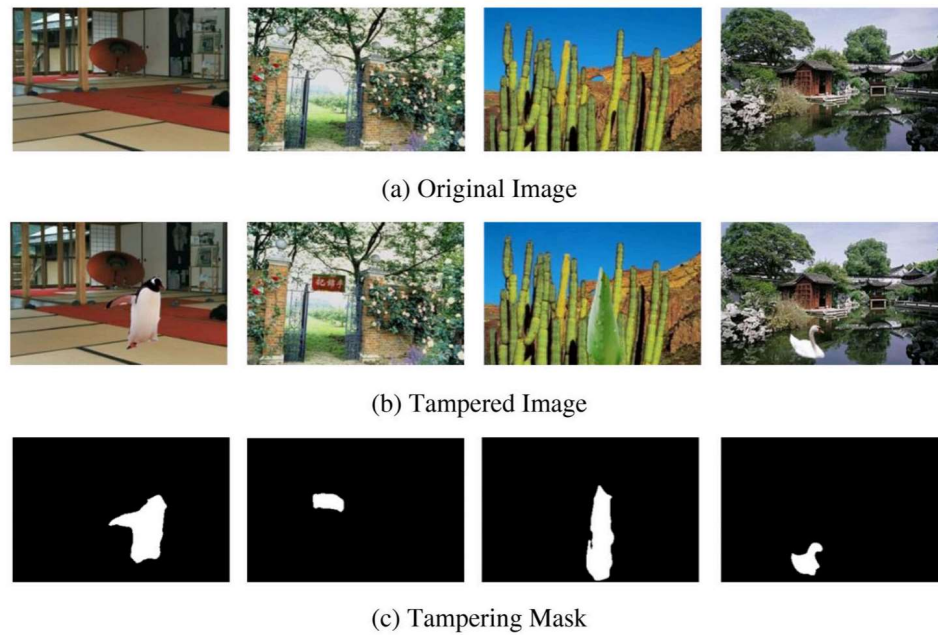


FIGURE 5 Detection results of our approach. (a) Original image, (b) tampered image, (c) tampering mask

TABLE 3 Ablation experimental results

Ablation model	F1-score	MCC-score
Base model	0.7232	0.6976
+SRM stream	0.7386	0.7051
+Self-attention	0.7481	0.7301
+ SRM stream and self-attention	0.7653	0.7447

3.6 | Robustness analysis

In order to verify the robustness of the algorithm, we conduct relevant post-processing operations on the tampered images in the test set. In detail, two types of operations that often used

to cover up tampering marks in images, namely JPEG compression and Gaussian smoothing, are used for interference. For the JPEG compression operation, different compression factors are used in the experiment, and they are 95, 90, 80, and 70. For the Gaussian smoothing operation, the size of the Gaussian kernel is 3×3 , and the standard deviations are 0.5, 1.0, 1.5, and 2.0, respectively.

In the experiment, the robustness of the model is measured by the F1-Score and MCC-Score indicators. The detection results after compression by different compression factors are shown in Table 4. It can be seen that as the compression factor decreases, the detection accuracy of the model shows a decrease trend. When the compression factor of the tampered image is 95, the model still has a high detection accuracy. However,

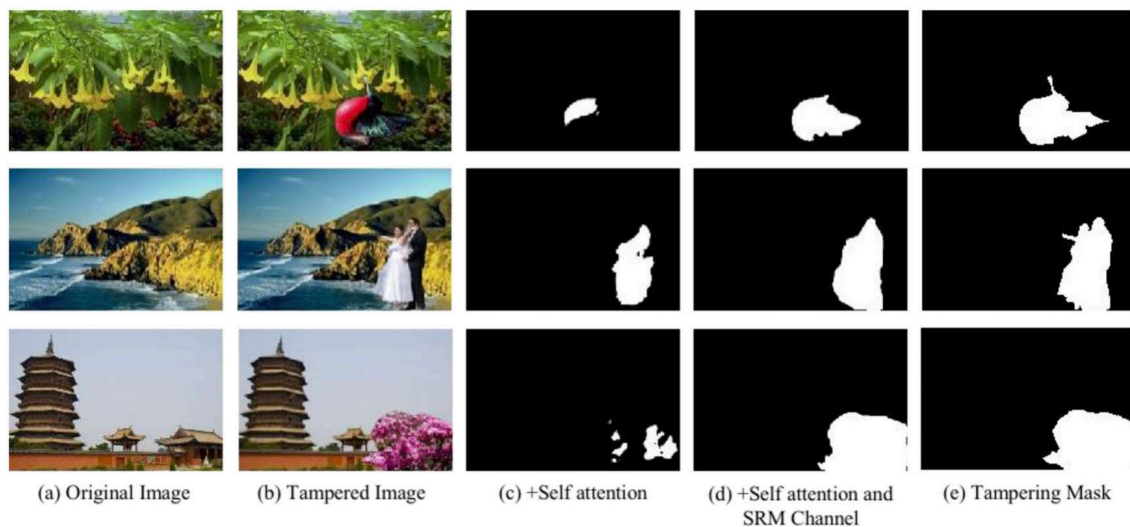


FIGURE 6 Detection results of ablation experiments

TABLE 4 Detection results after compression with different compression factors

Evaluation metrics/ compression factors	95	90	80	70
F1-score	0.6948	0.5461	0.1797	0.0995
MCC-score	0.6749	0.5376	0.2339	0.1785

TABLE 5 Detection results after Gaussian smoothing with different standard deviations

Evaluation metrics/ standard deviation	0.5	1.0	1.5	2.0
F1-score	0.6251	0.6338	0.6263	0.6229
MCC-score	0.6315	0.6136	0.5985	0.5923

when the compression factor is lower than 90, the detection accuracy of the model plummets. The reason is that the JPEG compression of the low compression factor causes the image information losses, and the model proposed in this paper is more sensitive to the detailed features of the image, so the detection accuracy is greatly reduced.

The detection results with the Gaussian kernel with different standard deviations as the interference method are shown in Table 5. It can be seen that as the standard deviation increases, the detection accuracy of the proposed model also tends to decline, but the anti-interference ability of the proposed lightweight model for Gaussian smoothing is very strong, for the F1-Score of the model still has 0.6229 when the standard deviation of Gaussian smoothing is 2.0.

4 | CONCLUSION

In this paper, a lightweight image splicing tampering localization algorithm based on MobileNetV2 and SRM is proposed. In order to reduce the complexity of the model and speed up the detection speed of the algorithm, this paper uses a lightweight convolutional neural network to reduce the multiple of the downsampling of the existing MobileNetV2 network. In addition, the model uses a dual-stream network, which uses the RGB stream to extract tampering features such as strong contrast differences and unnatural tampered boundaries, and uses the SRM noise stream to extract the noise inconsistency between image tampered areas and real areas. Moreover, the extracted dual-stream features also improve the sensitivity of the model to important features in RGB features and SRM features through an improved CBAM attention mechanism. The model still has high detection accuracy while achieving light weight, but it cannot identify tampered images after low compression factors. Our future work is to improve the robustness and detection accuracy of the model.

ACKNOWLEDGEMENT

This work was supported by Wuxi University Research Start-up Fund for Introduced Talents, and the Project of "Qing Lan Program" Excellent Teaching Team (No. 10 of Official Document from Jiangsu Education Department in 2020).

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in <http://forensics.idealtest.org/> and <http://www.ee.columbia.edu/ln/dvmm/downloads/AuthSplicedDataSet/AuthSplicedDataSet.htm>.

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How to cite this article: Shi, X., Li, P., Wu, H., Chen, Q., Zhu, H.: A lightweight image splicing tampering localization method based on MobileNetV2 and SRM. *IET Image Process.* 17, 1883–1892 (2023).
<https://doi.org/10.1049/ipr2.12763>